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Tweedle: Sensitivity Check in Health-related Social Short Texts based on Regret Theory

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Abstract

Twitter helps us to know what is happening in the world and what people are talking right now. Every day, millions of Twitteratians tweet something personal or impersonal to express their emotions and valuable knowledge. In the health domain, disclosure of personal health information will have a long-term effect to common individuals either directly or indirectly, which emphasize the presence of unrealistic social boundaries and the need of sensitivity analysis in social media. The proposed Tweedle framework was built with 100K tweets extracted based on a set of 20 health-related cyber-keywords. The framework of Tweedle was bounded with Regret Theory for tweet annotation, content and contextual feature scores for feature selection and various machine learning algorithms for sensitivity classification. The tweets annotated in accordance with Regret Theory by domain experts of Amazon Mechanical Turk resulted in 61.5% of sensitive tweets with health data. The context and content-oriented features scores are introduced in terms of Primary / Secondary tweet score, Named Entity Recognition Score of tweets, Term Frequency-Inverse Document Frequency (TF-IDF), Cyber-Keyword Ratio in tweets, hashtag mentions, user mentions as features for classification. The Tweedle experimented Regret Theory in combination with various classifiers like Support Vector Machine, Naïve Bayes, Random Forest, Decision Tree, Logistic Regression and Recurrent Neural Network + Long Short-Term Memory for sensitivity classification in health domain tweets. The training and testing results proved RNN + LSTM as the better performing model to identify tweets with sensitive health data.

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1. Introduction

Social media has a profound impact on how people communicate in the 21st century. Public communication has undergone a tectonic shift over the past decade with the rise of internet based social networking applications such as Facebook, Twitter, Snapchat, Instagram. These online social networks have unique features which make them attractive to various users wanting to communicate and spread information instantly. The ever-growing social networks present a unique opportunity to disseminate information by engaging, organizing and mobilizing people to address various alarming issues of recent times. For effective communication, the user must consider two important factors, namely the structure of their social network and the textual content published. The data published by users in online social networks are generally fuzzy and unstructured in nature and requires extensive processing rules to extract logical patterns and useful insights. The Figure 1 represents the cyber security threats that arise in the cyberspace in conjunction with social media.

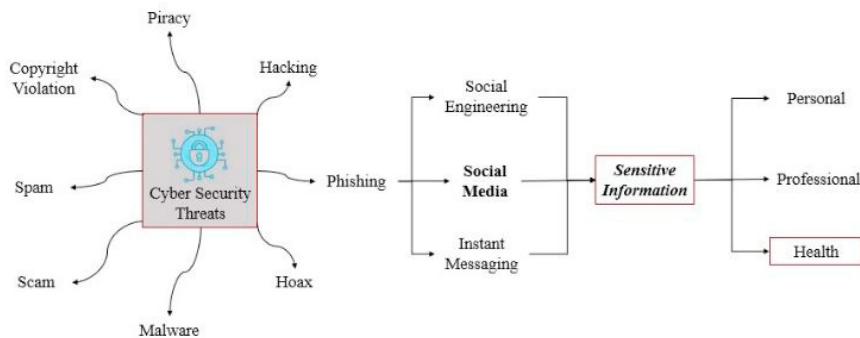


Fig. 1. Social media contribution to cyberspace for disclosure of personally identifiable information and its threats in various domains.

The privacy and self-disclosure might take a wide spectrum in the area of cyber security threats because it is directly related to trust and vulnerability of a person in a digital environment and their control over the personal data of self or other users [1, 15]. The experimental results proved that the computer mediated communication (CMC) and internet-based behaviour had higher volume and risks of regrets and self-disclosure when compared to face-to-face communication. Therefore, amongst various dimensions of privacy threats, the most critical and vulnerable issues arise from informational privacy threat which is influenced by an individual's right to determine what, when and how to share any personal information about self or others in a public forum. Though this relates to psychological factors, it has a higher dependence in other privacy threats across areas like financial, physical and professional bounds.

The National Institute of Standards and Technology (NIST) of the United States [2] and European Union's (EU) General Data Protection Regulation (GDPR) [3] defined any physically unique, psychologically expressive, cultural, social, biometric, genetic and health data as Sensitive Personal Data (SPD) or Personally Identifiable Information (PII). The presence of these PIIs in a social media post publicly will aid cyber criminals and stalkers to attack the users in physical, financial and psychological basis like mental depression, stress, anxiety, job loss, relationship loss and embarrassments.

The 28% of internet users are discussing about the health in online social networking sites like Facebook, Twitter, PatientsLikeMe. These data evidently lead to the research of identify and analyse Personal Health Information (PHI) in user posts [16]. Twitter does not provide any preventive signals to users when they include PHI in their tweet. Meanwhile, Twitter recommends four questions to their users to answer before posting any tweet. The privacy analysis questions included in Twitter's privacy policy are "Who am I sharing this information with?", "How much and what type of information am I sharing?", "How many people can see the information I am sharing?", "Can I trust all the people that see this information?". It is not possible to take the overall control of information diffusion and protect the user tweets but it is possible to technically leverage self-disclosures of PHI by learning the revealing scenarios and data types to take effective measures in user's privacy and regrets in OSN.

This research work aims at predicting the presence of personal health data in social media texts that includes narratives and suggestion on the health conditions of self or other users. The proposed framework Tweedle, computes scores based on a wide range of content and contextual features that were extracted from the raw tweets streamed with respect to a set of 20 health-related cyber-keywords. The Tweedle feature extraction process considers primary and secondary information disclosure in tweets based on the factor of how information leaks occur. It was broadly categorized into two kinds, namely primary information leak and secondary information leak [4]. The primary information leaks refer to posts with some sensitive PII's about the user himself / herself, whereas the secondary information leaks refer to the posts that contain some other user's sensitive PII's and its relevant data posted by the user. Similarly, various other factors of textual context like Named Entity Recognition (NER), hashtags, user mentions, cyber-keywords and content features like Term Frequency – Inverse Document Frequency (TF-IDF) were applied for building features for classification. The Tweedle framework experimented various classification models namely Support Vector Machine, Naïve Bayes, Random Forest, Decision Tree, Logistic Regression and Recurrent Neural Network + Long Short-Term Memory on features generated with Regret Theory. The experimental results showed wide deviations in classification with RNN+LSTM yielding greater accuracy than the other models.

The remainder of this paper is organized as follows. The Section 2 introduces the background and related works in privacy issues related to self-disclosures in social media and social media mining. The Section3 defines the proposed Tweedle framework. The Section4 presents experimentsandresult from analysis for the real-time data extracted for sensitivity analysis. Finally, the Section 5 briefs on the conclusion and future works in regard to sensitive health data in online social networks.

2. Literature Survey

The understanding of cyber security threats that occur just due to private data disclosure was explored with data from many social networks like Facebook, Twitter, Instagram, Flickr, Snapchat, etc., It is important to detect sensitive health data disclosure in Twitter because, it is one of the most frequently used platform where users share their health related experiences and practices [5]. The following sections briefs about various research works carried out inline with the scope of privacy leaks, health data in social media and social media text mining.

2.1. Privacy Leaks through Self-disclosure in Social Media Texts

The authors of [4] have contributed three methods for analysing the privacy leak in twitter. First, the authors characterized the nature of privacy leaks on Twitter to gain an understanding of what types of private information people are revealing on it. Second, using this characterization the authors built automatic classifiers to detect incriminating tweets for these three topics in real time in order to demonstrate the real threat posed to users by, e.g., Burglars and law enforcement. Third, they characterized who leaks information and how. The authors [1] have considered self-disclosure of private information in internet via various social media that may create negative consequences. The analysis was carried out by measuring the rate of privacy information disclosure in social media and explained the differences between privacy leak and self-disclosure.

The research work [6] presented a comparative analysis on the extent of user engagement in health conditions/issues on Twitter. Moreover, a measure over or under representation by comparing the tweet volumes with the corresponding prevalence of the health issues. The themes of public interest are measured by classifying tweets into several categories, namely, advice, awareness, news, statistics, advertisements and also, others, when it does not fit any other category. In the context of mental health, [7] reported that discourse on emotionally laden traumatic experiences can be a safe way of confronting mental illness. Disclosure has also been associated with reduced visits to medical centres and psychological benefits in the form of improved effect.

Self-disclosure has also been investigated as an outcome of alcohol intake, gender differences and personality characteristics under those circumstance internet-based behaviour by author [8] to analyse the self- disclosure of computer mediated communication. In the paper [9], the author made use of twitter since it may reflect attitudes represented in traditional surveys. This study aimed to understand professional's and the patient's thoughts regarding brachytherapy on Twitter. Another research work [17] was carried out to predict the presence of

medication intake in social media text posted by users of Twitter. It used supervised algorithms like SVM, Random Forest, Naïve Bayes, Logistic Regression and ensemble to classify tweets based on three level classes namely no intake, intake and possible intake.

2.2. Text Mining in Social Media Texts

In the paper [5] author exploited the wealth of user-generated data, available through the micro-blogging service Twitter, to estimate and track the incidence of health conditions in society. The method is based on two stages, extracting possibly relevant tweets using a set of specially crafted regular expressions, and then classify these initial messages using machine learning methods. In research work [10] mainly concerned with the text mining and analysis, including pre-processing which employed the natural language processing (NLP) techniques in social media to extract the subjectivity of the textual information on health-related tweets based on two steps: emojis labelling and semantic labelling based on text scoring. Each annotated dataset was evaluated and then fed into sentiment prediction models to classify tweets. Similarly, another research work was carried out to detect PHI in tweets. The PHI ontology was extracted and evaluated with Naïve Bayes classifier with health-related semantics [16] to analyse and detect tweets with PHI.

There are various deep learning models and methods evolved in the context of NLP and social media monitoring [15] to overcome the limitations of conventional machine learning models. There are many traditional NLP techniques that dominate the computational task of text features like parts-of-speech (POS) tagging, NER, Lexicon analysis [11]. The drawbacks in traditional machine learning approaches were solved by using deep learning models which enable multi-level feature extraction automatically with reduced time complexity. A deep contextualized word representation was proposed by [12] that used complex characteristics of word usage and linguistic contexts using the deep bidirectional language model (biLM). This approach was helpful in improving the handling the NLP challenges faced during sentiment analysis, identifying questions and answering patterns.

The process of extracting health related information from electronic health records was performed by [13] using natural language processing techniques like Named Entity Recognition (NER). The health data were identified using three different approaches, namely, Exact NER, Fuzzy NER and Lemmatized or Stemmed NER. The other contextual features like the co-occurring words and contextual importance of a word were not considered for analysis. The short text classification was also performed by combining lexical and semantic features using topic detection [18, 19] which proved better performance when compared to existing methods. The feature selection for short text was improved by [20] with a modified genetic algorithm which faced difficulty in handling the spatiality of short texts of online social networks.

3. Methodology

The proposed framework of Tweedle system will be used for sensitivity check in disclosure of PHI that helps to avoid negative consequences. The Figure 2 shows the architecture of Tweedle framework. Tweedle adapted the strategy of the Regret theory to identify contextual features from tweet during annotation process and to compute various scores which proved to be highly influential in predicting sensitive tweets. The proposed strategy of using both regret scores with TF-IDF scores was better performing in identifying sensitive tweets with PHI when compared to other traditional techniques used for short text classification [21].

Regret Theory – According to the social media environment, regret theory may be defined as, “*the decisive mode of the users to decide on what to post, rethink on the posting content and make sure they don’t regret for the post in future*”. This cognitive state is supported by comparing the tweet and user state with strategical tweet regret types defined by twitter users – blunder, direct attack, group reference, reveal or explain too much, direct or implied criticism, expressive, lie, agreement changed or withdrawn, behavioural edict [26, 27].

This perspective was emphasised as a rule during tweet annotation performed by domain experts of AMT. To contribute in the decision making, Tweedle uses the strategy of combining contextual and content features of a tweet. The framework of Tweedle is explained in the following sections.

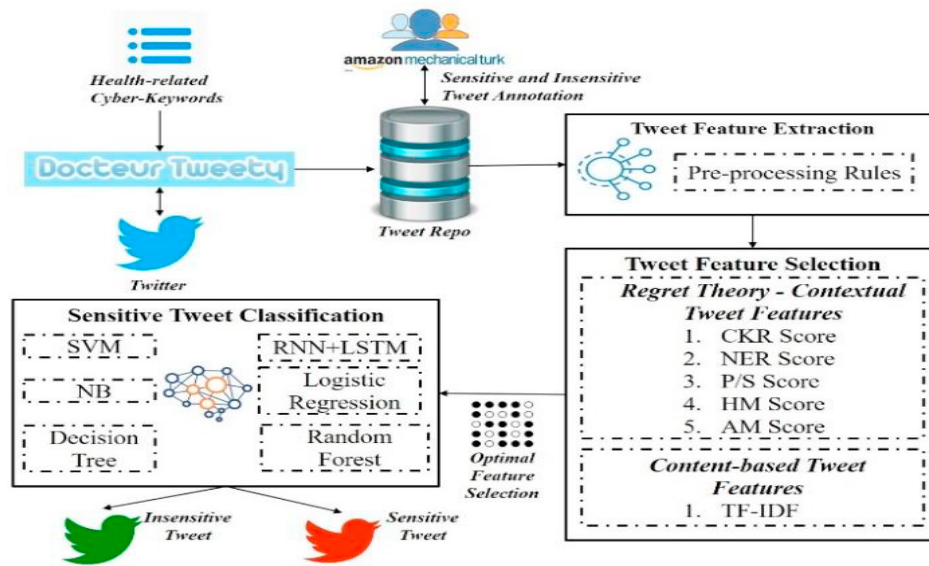


Fig. 2. Proposed Tweedle Framework using Regret Theory.

3.1. Twitter Data Collection

The tweets are collected from Twitter using Docteur Tweety, a streaming online interface for extracting raw tweets based on certain query terms. Totally, 2,34,088 tweets are collected using 20 different health-related cyber-keywords from 10/2018 to 21/11/2018. The tweets to be selected for analysis was determined by filtering news, advertisements, spams, entertainment and reviews so that the dataset will be well suitable for predicting health-based sensitive tweets. The annotation of tweets for the presence of SPD was performed by a third-party service provided by AMT. The domain experts of AMT were provided with rules and assumptions that was framed in accordance with the privacy policies of various standardized institutions [2,3] and regret theory. The regret theory for the users of online social networks can be defined as the mental state that they experience after posting any form of content in any social media. It is good if the users don't regret or feel sad or disturbed after posting any personal information and later claim it as a wrong / regretful activity.

3.2. Tweet Pre-processing

The tweets collected are generally unstructured, inconsistent and noisy, which might produce misleading mining and classification results. Data pre-processing is a proven method of resolving such issues. It transforms raw data into a machine-readable format. It also filters out useless data. The various steps involved in it are data cleaning, integration, reduction, and transformation are recommended to achieve optimal performance in classification [21]. The list of pre-processing rules applied is shown in Table 1.

Table 1. A list of pre-processing rules applied on a tweet.

Rule No.	Pre-processing Rule	Rule Description
1	Convert to lower case	Transform tweet to small letters
2	RT removal	Remove retweets (RT)
3	Replacement of user mentions	Replace @ in tweets with 'NAME'
4	URL Replacements	Replace URL links with 'URL'
5	Person name removal	Stanford and Twitter Named Entity Recognizer was used to remove person names from tweets

Rule No.	Pre-processing Rule	Rule Description
6	Hash character removal	Remove '#' in hashtags.
7	Removal of punctuations and symbols	Remove punctuations and symbols, including images and non-English characters
8	Removal of smiley	Remove various smiley that expresses tweet's emotion
9	Lemmatization	Stanford Core NLP was used to convert words to base form
10	Stop-word Removal	English stop-word list provided by NLTK was used to remove stop words like a, the, and, etc.,
11	Stripping Whitespaces	Remove additional whitespaces

3.3. Contextual Feature Extraction

The contextual features are extracted from the textual content of the tweets. The list of content features is shown in Table 2. Each contextual feature is computed for all tweets individually. The blending of all the five-feature score is termed as Regret Score for a particular tweet. This score will be used to cross check the degree of vulnerability a tweet will pose when posted in public.

Table 2. Contextual Features of Tweedle.

Feature	Feature Description
CF1	Cyber Keyword Ratio (CKR)
CF2	Named Entity Recognition (NER) Score
CF3	Primary or Secondary (P/S) Identification
CF4	'#' Hashtag Mentions (HM) Score
CF5	'@' User Mentions (AM) Score

3.3.1. Cyber-Keyword Ratio (CKR) Score

To find the privacy leak of health domain tweets, Tweedle has used set of 20 cyber-keywords related to the health domain like accident, allergy, immunity etc. CKR is a proposed feature that computes the ratio between the number of cyber-keywords (CK) within a single tweet to the total number of cyber keywords (CK) chosen for collecting the overall dataset.

$$CKR = \frac{\sum_i CK_i}{\sum_n CK_n} \quad (1)$$

where, i denotes the number of cyber-keywords within the tweet, n denotes the total number of cyber-keywords in health domain.

3.3.2. Named Entity Recognition (NER) Score

NER is a subtask of information extraction that attempts to locate and classify the real time entity that are mentioned in unstructured text. Twitter NER has few pre-defined categories such as the person names, organizations, locations, medical codes, time, quantities, monetary values, percentages, etc., Thus, based on the presence of entity values, NER score was computed as shown in Equation. 2.

$$NER = \frac{\sum_i E_i}{\sum_r E_r} \quad (2)$$

where, i denotes the number of entities within the tweet, r denotes the total number of entities present in the corpus.

3.3.3. Primary / Secondary (P/S) Tweet Score

POS tagging [14] is the process of marking words in a corpus to a corresponding part-of-speech tag, based on its context and definition. This task is not straightforward, as a particular word may have a different part of speech based on the context in which the word is used. The Tweedle uses POS tagging approach for identification of

Primary /Secondary tweet by comparing the presence of pronoun (PRP) and its various kinds. As defined earlier, the primary and secondary tweets are marked based on the concept of primary and secondary information leaks [4] as shown in Table 3.

Table 3. P/S Scores based on POS Tags

Primary / Secondary Tweet Identification	Score
Contains only primary pronouns(I, My, Me, Am, etc.,)	1
Contains only secondary pronouns (He, She, They etc.)	2
Contains both primary & secondary terms	3
Does not contain any pronouns	0

3.3.4. '#' Hashtag Mention (HM) Score

Hashtags are a great way for people to search for topics on social networks that have a common theme. They are also used to signpost a conversation topic. For example, any health organization shares the new disease spread through hashtags like #SwineFlu create awareness among public basically to alert. The importance of Hashtags is competition, branding and promotion, visibility, activism, recap. Here activism in health domain has been analysed to find the hash mention by taking the number of hashtags (HM) in every tweet as depicted in the following Equation 3.

$$HM = \sum_i HM_i \quad (3)$$

where, i denotes the number of hashtags mentioned by the user within the tweet.

3.3.5. '@' User Mention (AM) Score

In @ User Mentions (AM), the user will either refer someone while sharing a tweet on twitter or will be referred by someone that may sometimes contain private information that may be a cause for negative consequences. So Tweedle proposes a feature called AM score, which will be computed for every health domain tweet as shown in Equation 4.

$$AM = \sum_i AM_i \quad (4)$$

where, i denotes the number of users mentioned by the user within the tweet.

3.4. Content Feature Extraction

In addition to contextual features, the Tweedle framework incorporated Term Frequency-Inverse Document Frequency approach for feature extraction. The TF-IDF score is a statistical measure often used in information retrieval and text mining for computing the significance of a feature in the whole dataset which here relates to the significance of a word and its importance in the corpus. The importance increases proportionally to the number of times a word appears in the document, but is offset by the frequency of the word in the corpus. This TF-IDF is considered as content related feature that explains the term relation of a document and provides better understanding of tweet along with content feature for sensitivity check.

4. Results and Discussions

4.1. Data Collection and Organization

Tweets were collected for the month of October and November in 2018. 2,56,158 tweets were collected for set of cyber-keywords as presented in Figure 3 from Docteur Tweety. After eliminating retweets, the repository had 79,850 tweets. Since the Tweedle Sensitivity Check System is going to handle only self-disclosure of health-related tweets, the author considered only public tweets with less than 500 followers. Then by further eliminating replicative tweets, database with 11647 tweets was used for analysis. The Table 4 depicts the statistical data of

tweets collected. Out of 11647 tweets some of them were insensitive and others are sensitive as depicted in Figure 3 which was manually annotated according to Regret Theory by experts at Amazon Mechanical Turk (AMT) Service for every cyber-keyword.

Table 4. Dataset Description.

S.No.	Dataset Description	
1	Number of tweets collected	2,34,088
2	Number of cyber-keywords chosen	20
3	Time of data collection (t)	10/2018 to 11/2018
4	Data collection tool used	Docteur Tweety
5	Tweets considered for sensitivity prediction	11,647

The AMT workers were selected based on certain criteria like they should be aware of health factors, should be a regular user of social media – atleast one hour per day, should possess good English knowledge, etc., The workers were given details about the cyber-keywords, aim of sensitivity analysis and class categories (Sensitive and Insensitive), Regret Theory with examples to make sure the annotation process to be completed as per the Tweedle’s requirement. On analyzing the annotation results received from the expert workers of AMT, the distribution of sensitive and insensitive tweets for each health-related cyber-keyword was found to vary widely. The main challenge of analyzing real time social media data on a specific problem will be in handling the unbalanced dataset. The same problem of the unbalanced class is prevailing in Tweedle dataset which will be compromised by using the cross-validation approach during the training and testing phase, as discussed in section 4.5. The Figure 3 clearly presents the Sensitive and Insensitive tweets in each cyber keyword that represents the need for Sensitivity Check in the health domain tweet due to disclosure of private information.

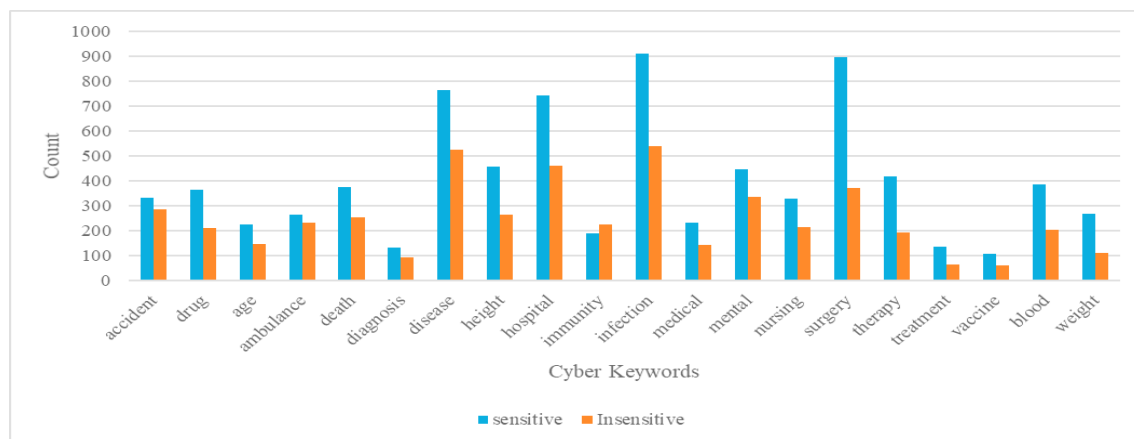


Fig. 3. Number of Sensitive and Insensitive tweets identified during the annotation process of AMT.

4.2. Pre-processing Data

A sample raw tweet is displayed in the Table 5. For this particular tweet, all the pre-processing rules described in the Table 1 are applied. The pre-processed tweet is displayed after performing each pre-processing rule over the raw tweet. The pre-processing rules should be executed in the same order in which they are described in the Table 1.

Table 5. A sample tweet instance before and after various transformations of the data pre-processing stage.

Tweet Instance	RT Almost got run over by an AMBULANCE whilst:) I was standing on the pavement!! Have a Look @ambulance #fear #irony“
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Pre-processed Tweet almost got run ambulance whilst stand pavement look URL NAME fear irony

4.3. Contextual Feature Extraction and Scores

The contextual features are extracted from the raw tweets because after the pre-processing stage, the tweet will lose its original content and structure. As mentioned earlier, CKR, NER, PRP, HM and AM scores will be calculated for every tweet with health domain data and stored with the corresponding tweet in the vector form. A set of sample tweet instances with contextual feature scores is presented in following Table 6.

Table 6. Contextual score computation for tweet instances.

Tweet	Cyber-keyword	CKR Score	Entities within Tweet	NER Score	PRP	P/S Score	Hashtag	HM Score	@ Mentions	AM Score
@DonaFrank18 Hello Sparky, how's it going. I found you here by accident. Hope you are doing well.	accident	0.04	None	0	i. you	3	None	0	@DonaFrank18	1
RIP FOR FIVE NATIONAL AND INTERNATIONAL LEVEL WEIGHT LIFTERS WHO DIED IN AN ROAD ACCIDENT IN NEW DELHI	weight, accident	0.08	1.Five-Cardinal 2.New Delhi-GPE	0.4	None	0	None	0	None	0
love is when you do not know when your friend's birthday, but you know the height and weight of the @FinnSkata	height, weight	0.08	-	-	you	2	None	0	@FinnSkata	1
#SadhakaDami airnewsalerts: Karnataka: 5 people killed, 8 injured in road accident in a village in Sibhara village in #Chitradurga district.	accident, injured, kill	0.14	1. SadhakaDami-person 2,3.karnataka,Sibhara-GPE 4,5. 5,8 -cardinal	1.2	-	0	SadhaDami, Chitradurga	2	None	0
@nxwestmidlands Today my Fiancee has had an accident on YouTube bus and get injured, where can i claim it to get compensation?	Accident, injured	0.08	1.Today- Date 2.Fiancee-Person 3.Youtube - Organisation	0.6	My, I	2	None	0	nxwestmidlands	1

4.4. TF-IDF Score

The TF-IDF matrix is one of the traditional ways of computing content-based features in text classification [21]. Since social media texts are unstructured and noisy, it usually generates a sparse feature set that needs some optimization for better classification results. The main challenge of TF-IDF matrix score is that it generates scores for all elements present in the pre-processed tweet. It cannot combine similar words and assign a common weight or score and pave way to feature reduction. Therefore, each word is independently associated with the dataset leading higher spatial features for larger datasets. Therefore, the optimal features are extracted by repeatedly testing the classifiers for varied number of feature set size by ranking the TF-IDF scores and varying the number of features in the range of 100, 200, ..., 10,000 for the Tweedle dataset.

4.5. Sensitive Tweet Classification

The Tweedle framework was tested with four state-of-art classifiers (SVM, Naïve Bayes, Decision Tree, Random forest), one regression (Logistic Regression) and Recurrent Neural Network with LSTM algorithms with the proposed content and contextual features. The evaluation metrics such as true positive, false positive, true negative, false negative can be found by computing the values of accuracy, precision, recall, F1 score. The unbalanced class property of Tweedle dataset was rectified by applying 10-fold cross validation mechanism during the training phase. The Tweedle framework was built by using 80% of tweets in training and 20% of tweets to test the trained machine learning models.

The Table 7 and Table 8 shows performance metrics of Tweedle framework. The RNN+LSTM model performs better for our content and contextual features in top 500 TF-IDF. Since the dataset available is sparse, the top 500 and top 5000 values from TF-IDF matrix was extracted during the feature selection process. The accuracy can be contributed by only large number of True Negatives, but F1 Score might be a better measure to use, to seek a balance between Precision and Recall, since our tweets exists as an uneven class distribution (large number of Actual Negatives).

Table 7. Evaluation metrics for top 500 TF-IDF with content and contextual features

Classification Algorithm	Accuracy	Precision	Recall	F1 Score
Naïve Bayes	50.01	62.27	43.39	51.56
SVM	62.63	62.59	99.40	76.81
Decision Tree	61.71	62.18	52.34	57.26
Random Forest	58.41	62.62	79.38	70.17
Logistic Regression	59.27	61.67	85.32	71.73
RNN+LSTM	67.06	75.06	93.11	83.11

On analysing the performance metrics, the deep learning model with RNN and LSTM outperformed all the state-of-art algorithms due to its recursive weight computation for each feature vector. The sequential processing of RNN helps in tracking the contextual features like the co-occurring words, identifying similar words and incorporate syntactic word representation. This highly brings claimable spatial reduction and context dependency in tweet features. As the size of the feature vectors increases, the sparsity increases and the accuracy seem to deteriorate in all the classifiers.

Table 8. Evaluation metrics for top 5000 TF-IDF with content and contextual features.

Classification Algorithm	Accuracy	Precision	Recall	F1 Score
Naïve Bayes	46.40	61.70	29.85	40.52
SVM	59.97	61.80	90.33	73.44
Decision Tree	61.71	95.01	75.12	76.01
Random Forest	58.45	61.01	88.24	72.15
Logistic Regression	53.17	62.59	60.15	61.40
RNN+LSTM	64.09	69.09	87.14	77.07

The Tweedle framework also utilizes content and contextual features in classification which not performed in the existing works of sensitivity analysis. The following Table 9 will depict the difference and significance of Tweedle against other existing works. The factor of using Regret theory is to narrow down the varying perspectives of privacy between various users in defining what is sensitive. This was achievable by Tweedle by proposing rules and recommendations to AMT expert workers in annotation tweets without the intervention of the user who posted the tweet. This helps the RNN to learn the various dynamics and degree of sensitive PHI in tweets and predict the vulnerability in future.

Table 9. Comparison of Tweedle framework with Regret Theory with the existing methods in determining PHI in Tweets.

Research Work	Approach	Classification Methods	Dataset - Inference	Score Computation
Using Transfer Learning to Identify Privacy Leaks in Tweets (2016) [22]	Transfer Learning – Spectral Clustering Method, SVD, PCA	AdaBoost, KNN, Random Forest, J48, SVM, K-means	1. Relative labelling of new tweets by differentiating words as domain independent words and domain-specific words. 2. Identification of privacy leaks in pregnant and drunk data	N/A
A content analysis of depression-related tweets (2015) [23]	Manual coding and annotation process	Annotator's knowledge base	1. Dataset - Random sample of depression related tweets. 2. More than 66% of tweets revealed Major Depressive Disorder.	Tweets filtered based on Klout Score [28]
Are health-related tweets evidence based? Review and analysis of health-related tweets on twitter. (2015) [24]	Evaluation by American Board – medical consultant.	Based on popularity, activity, interaction and frequency of tweets.	1. 25 Twitter accounts (physician, nonofficial health institutes, dietitian, and government institute) – 625 Tweets 2. More than 50% of information posted by professionals were false as per the expert review. 3. Tweets posted by physicians were true.	N/A
Privacy detective: Detecting private information and collective privacy behavior in a large social network. (2014) [25]	Topic Modelling, Named Entity Recognition, Privacy Ontology, Text Normalization, and Sentiment Analysis.	AdaBoost with Naïve Bayes as weak learner	1. 500,000 tweets from 100,000 Twitter users. 2. Tweets of 270 users were annotated for 9 categories by AMT workers – Private or not? 3. Privacy score of a user and their correlation with user friend's score, mentioned user's privacy score.	Privacy Score of a user 1. Privacy Score 1 - >70% of tweets non-private. 2. Privacy Score 2 – 30% - 60% of tweets private. 3. Privacy Score 3 - >60% of tweets private. Perplexity scores to determine the tweet topic.
Proposed Work - Tweedle framework with Regret theory	Cyber-Keywords, Named Entity Recognition, Hashtags, User Mentions, Primary / Secondary Tweet	Naïve Bayes, SVM, Decision Tree, Random Forest, Logistic Regression and RNN+LSTM Regret Theory	1. 2,34,088 tweets extracted from 20 different health-related cyber-keywords from 10/2018 to 211/2018 2. Tweets annotated by AMT workers based on Regret Theory 3. RNN+LSTM performed better in learning tweets with PHI 4. Regret Theory was efficient in determining the PHI due to the nature of varying privacy perspectives among users.	Tweet specific Score – Content and Contextual 1. Cyber Keywords Ratio 2. NER Score 3. Primary / Secondary Score 4. '#' Hashtag Score 5. '@' User Mention Score 6. TF-IDF Score Regret score can be inferred by computing the average of all the scores associated with the tweet.

Finally, the regret score for a tweet can be analysed to check the degree of vulnerability a tweet possesses when posted. Therefore, from the scores computed in Tweedle framework, a user can check a tweet's sensitivity state by computing the average of CKR, NER, HM, P/S and AM scores. The authors recommend the users to be aware of the presence of cyber-keywords, data in each and every real-world entity, self or secondary disclosures, users mentioned in a post to avoid regrets in online society.

5. Conclusion and Future Works

Disclosure of private information increases day by day in social media with the potential threats and vulnerabilities going unnoticed by users. Hence, precautionary steps must be taken before getting into negative consequences. The proposed Tweedle framework helps to check the sensitivity of the social media texts in the health domain by applying regret theory over the textual content. The major contribution of this project is to check the sensitivity of the tweet using special features like Named Entity Recognition (NER), Parts of Speech (POS) Tagging, Cyber Keyword Ratio (CKR), Hashtag Mentions (HM) and At Mentions (AM) along with TF-IDF feature to obtain better results. From the obtained results, the system performs better with all those features in health domain. The RNN + LSTM is found to be the best algorithm respectively with 83.11 accurate sensitive health tweet classification.

Though there are many privacy settings available on social networking sites, the users must be aware of using it in the right way. The reality of cyberspace is that the users are unknown of a potential audience of their posters. Since the number of active users and way of communication through internet increases, leakage of private information will continue to happen in social media in various forms. The Tweedle with the deep recurrent model will help in identifying the self-disclosure and random disclosures of health-related data on social networking sites. In the future, the proposed framework can be enhanced with the feature of user preference in identifying sensitive data. The disclosure of health-related tweets was only taken in Tweedle system; furthermore, domains can be taken into account. The system can also be enhanced to include support for varied kinds of pre-processing techniques and mining such as image and video mining.

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